

These tasks progress from simple arithmetic to practical real-world examples, reinforcing the concept that a **function performs a calculation and returns a value that can be stored, reused, or combined with other calculations.**

Learning Progression

Task	Skill Practiced	Returns
1	Add numbers	Integer
2	Multiply numbers	Integer
3	Rectangle area	Integer
4	Employee pay	Integer
5	Student average	Float
6	Time conversion	Integer
7	Temperature conversion	Float
8	Discount calculation	Integer
9	Simple interest	Float
10	Accounting equation	Integer

Python Value-Returning Functions – Progressive Practice Exercises

Each task contains **three exercises** that gradually increase in difficulty.

Difficulty Levels

- Exercise 1 – Beginner (Direct Function Call)
- Exercise 2 – Intermediate (User Input)
- Exercise 3 – Advanced (Multiple Function Calls or Extra Calculations)

Python Value-Returning Functions – 10 Practice Tasks

These exercises help students understand that a **function returns a value** instead of just printing it. Each task follows the same learning pattern:

1. Define the function.
2. Pass arguments.
3. Return a value.
4. Store the returned value in a variable.
5. Print the result.

Task 1 – Add Two Numbers

Goal

Create a function that returns the sum of two numbers.

 Python

```
def add_numbers(num1, num2):  
    return num1 + num2  
  
answer = add_numbers(15, 25)  
  
print("Answer:", answer)
```

Expected Output

Answer: 40

Task 1 – Add Two Numbers

Exercise 1

Create a function named `add_numbers()` that returns the sum of **18** and **42**.

Expected Output

Answer = 60

Exercise 2

Ask the user to enter two numbers.

Pass the numbers into the function.

Store the returned value in a variable named `total`.

Display:

The total is: ____

Exercise 3

Create a function that adds **three numbers**.

Example

```
Enter number 1: 15  
Enter number 2: 25  
Enter number 3: 40  
  
Total = 80
```

Task 2 – Multiply Two Numbers

Exercise 1

Return the product of:

12 x 7

Task 2 – Multiply Two Numbers

Goal

Return the product of two numbers.

</> Python

```
def multiply_numbers(a, b):  
    return a * b  
  
result = multiply_numbers(8, 6)  
  
print("Product:", result)
```

Expected Output

Product: 48

Task 2 – Multiply Two Numbers

Exercise 1

Return the product of:

12 x 7

Exercise 2

Ask the user for:

- Width
- Height

Use a multiplication function to calculate the area.

Exercise 3

Create a program that calculates the volume of a box.

Formula

Volume = Length x Width x Height

Return the volume from the function.

Task 3 – Find the Area of a Rectangle

Exercise 1

Find the area of

Length = 9
Width = 6

Task 3 – Find the Area of a Rectangle

Formula

Area = Length x Width

Python

```
def rectangle_area(length, width):  
    return length * width  
  
area = rectangle_area(12, 5)  
  
print("Area:", area)
```

Expected Output

Area: 60

Exercise 1

Find the area of

Length = 9
Width = 6

Exercise 2

Ask the user for the rectangle dimensions.

Return the area.

Display

The rectangle area is ____ square units.

Exercise 3

Calculate the total area of **three different rectangles**.

Example

Rectangle 1 Area = ____

Rectangle 2 Area = ____

Rectangle 3 Area = ____

Total Area = ____

Task 4 – Calculate Employee Weekly Pay

Formula

Pay = Hours × Rate

Python

```
def weekly_pay(hours, rate):  
    return hours * rate
```

```
pay = weekly_pay(40, 22)
```

```
print("Weekly Pay: $", pay)
```

Expected Output

Weekly Pay: \$ 880

Exercise 1

Employee worked:

40 hours

\$22 per hour

Return weekly pay.

Exercise 2

Ask the user for:

- Employee Name
- Hours Worked
- Hourly Rate

Display

Employee: John

Weekly Pay = \$____

Exercise 3

Calculate the weekly pay for **three employees**.

Display

Employee A

Employee B

Employee C

Grand Payroll = \$____

Task 5 – Calculate Student Average

Formula

$$\text{Average} = (\text{Score1} + \text{Score2} + \text{Score3}) / 3$$

Python

```
def average_score(score1, score2, score3):  
    return (score1 + score2 + score3) / 3  
  
average = average_score(85, 90, 95)  
  
print("Average:", average)
```

Expected Output

Average: 90.0

Exercise 1

Return the average of

80

90

100

Exercise 2

Ask the user for **five subject scores**.

Return the average.

Exercise 3

Display both the average and grade.

Example

Average = 87

Grade = B

Grading Scale

90-100 A

80-89 B

70-79 C

60-69 D

Below 60 F

Task 6 – Convert Hours to Minutes

Formula

Minutes = Hours x 60

Python

```
def hours_to_minutes(hours):  
    return hours * 60  
  
minutes = hours_to_minutes(3)  
  
print("Minutes:", minutes)
```

Expected Output

Minutes: 180

Task 6 – Convert Hours to Minutes

Exercise 1

Convert

5 hours

into minutes.

Exercise 2

Ask the user to enter hours.

Display

Hours

Minutes

Exercise 3

Return

- Minutes
- Seconds

Example

Hours = 2

Minutes = 120

Seconds = 7200

Task 7 – Convert Celsius to Fahrenheit

Formula

$$F = (C \times 9 / 5) + 32$$

</> Python

```
def celsius_to_fahrenheit(celsius):  
    return (celsius * 9 / 5) + 32  
  
temperature = celsius_to_fahrenheit(25)  
  
print("Temperature:", temperature)
```

Expected Output

Temperature: 77.0

Task 7 – Convert Celsius to Fahrenheit

Exercise 1

Convert

30°C

to Fahrenheit.

Exercise 2

Allow the user to enter any Celsius temperature.

Display both temperatures.

Exercise 3

Determine whether the temperature is

Freezing

Cold

Warm

Hot

Example

20°C

68°F

Weather: Warm

Task 8 – Calculate Discounted Price

Formula

Final Price = Price - Discount

</> Python

```
def discount_price(price, discount):  
    return price - discount
```

```
final_price = discount_price(150, 20)
```

```
print("Final Price: $", final_price)
```

Expected Output

Final Price: \$ 130

Task 8 – Calculate Discounted Price

Exercise 1

Return the final price.

Price = \$200

Discount = \$35

Exercise 2

Ask the user for

- Product Name
- Original Price
- Discount

Display

Laptop

Original Price

Discount

Final Price

Exercise 3

Calculate the bill for **four products**.

Display

Subtotal

Total Discount

Final Total

Task 9 – Calculate Simple Interest

Formula

$$\text{Interest} = (\text{Principal} \times \text{Rate} \times \text{Time}) / 100$$

Python

```
def simple_interest(principal, rate, time):  
    return (principal * rate * time) / 100  
  
interest = simple_interest(5000, 6, 3)  
  
print("Interest:", interest)
```

Expected Output

Interest: 900.0

Exercise 1

Return the interest.

Principal = 4000

Rate = 5%

Time = 4 years

Exercise 2

Ask the user for

- Principal
- Rate
- Time

Return the interest.

Exercise 3

Return

Interest

Final Amount

Formula

$$\text{Final Amount} = \text{Principal} + \text{Interest}$$

Display

Interest

Final Amount

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Task 10 – Calculate the Accounting Equation

Formula

Assets = Liabilities + Equity

Python

```
def calculate_assets(liabilities, equity):  
    return liabilities + equity  
  
assets = calculate_assets(25000, 75000)  
  
print("Assets:", assets)
```

Expected Output

Assets: 100000

Exercise 1

Calculate

Liabilities = 40,000

Equity = 60,000

Return Assets.

Exercise 2

Ask the user to enter

- Company Name
- Liabilities
- Equity

Return Assets.

Display

Company

Assets

Liabilities

Equity

Exercise 3

Create a mini accounting report for **three companies**.

Display

Company A

Assets

Company B

Assets

Company C

Assets

Combined Assets

